

Credit Scoring Model: LR vs LightGBM — Production Decision

Champion-challenger readout · operating point: 5% FPR · offline evaluation on a held-out window

RECOMMENDATION: SHIP challenger

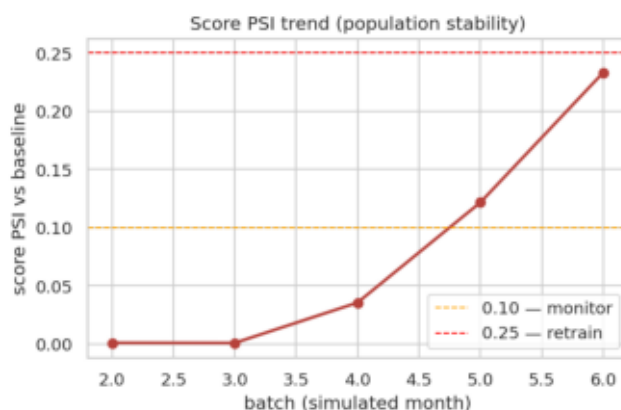
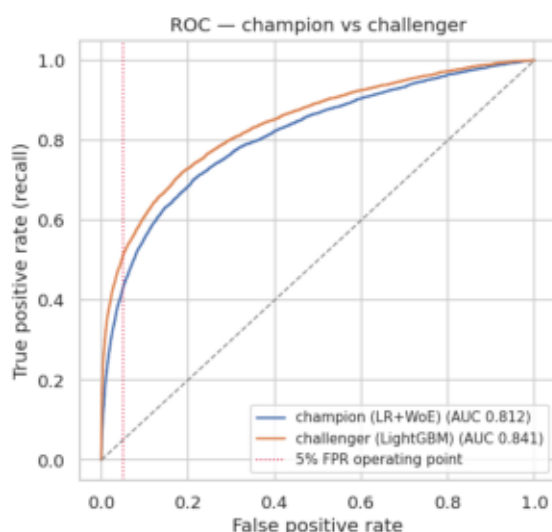
1 · Lift at the business threshold

At a 5% FPR operating point, challenger recall 51.1% vs champion 43.0% = +8.1pp more defaulters caught for the same share of good customers declined (95% CI [6.9, 9.3]pp, paired bootstrap, n=1000).

Discrimination — champion AUC 0.812 / KS 0.489 · challenger AUC 0.841 / KS 0.533. Rule (pre-registered): ship if lift $\geq$ 3pp AND CI lower $>$ 0 $\rightarrow$ met.

2 · Risk / cost trade-off

Extra defaulters caught (test window): 408 · loss avoided \$2,040,000 (assumes \$5,000/default). False-positive cost $\approx$ \$0 — both models compared at an identical 5% FPR, so the same share of good customers is declined. Net $\approx$ \$2,040,000 per window.



3 · Caveats

- Leakage screen: feature(s) with $IV > 0.5$ = RevolvingUtilizationOfUnsecuredLines; reviewed as the canonical strongest credit driver (revolving utilization), retained — not target leakage.
- Challenger (LightGBM) less interpretable than the WoE scorecard; isotonic calibration applied (see calibration chart).
- Training window is simulated batches; production needs a real vintage.
- Post-deployment monitoring is live: injected-drift sanity check PASSED (CSI caught DebtRatio drift, control age stable).

4 · Next step

Ship challenger behind the existing WoE scorecard as a shadow/A-B test; keep champion as fallback. Retrain trigger wired to PSI: investigate at 0.10, retrain at 0.25 (DebtRatio already breached in the drift simulation).